Trecode: a FAIR eco-system for the analysis and archiving of omics data in a combined diagnostic and research setting

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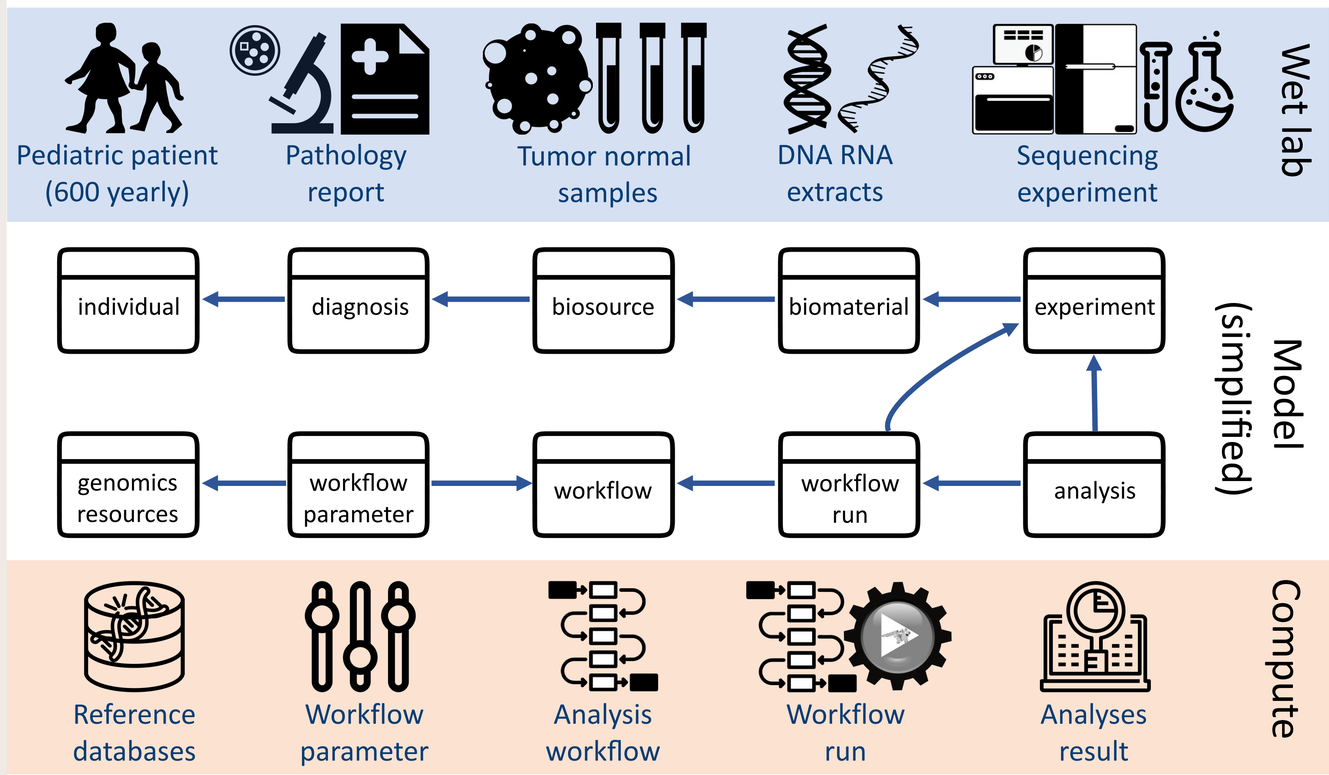
Introduction

Recently we released Trecode[1], a framework that records both clinical and research sample (meta) data and manages computational genome analysis workflows. In essence the framework ensures data provenance, genome analyses consistency and that data remain Findable, Accessible, Interoperable, and Reusable (FAIR).

Trecode is intentionally composed of compute backend agnostic components, which currently allows us to develop analyzes scalability by using cloud compute resources on an on demand basis.

The Trecode data model, codebooks, NGS workflows and client programs are publicly available. The complete software stack is coded in Ansible to facilitate automated deployment by others.

Biobank genomics captured in a single data model

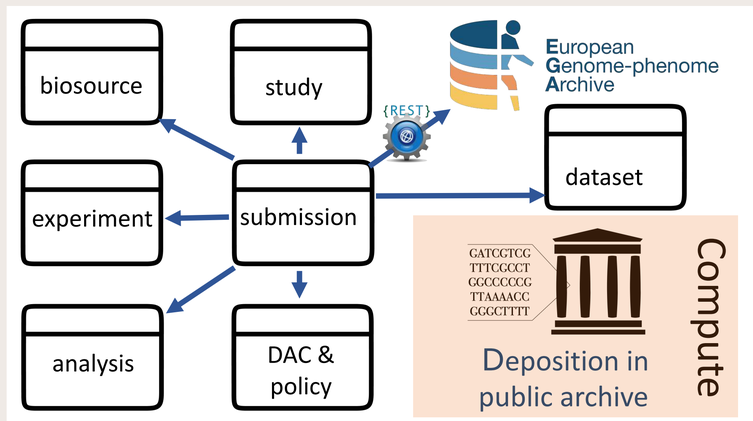


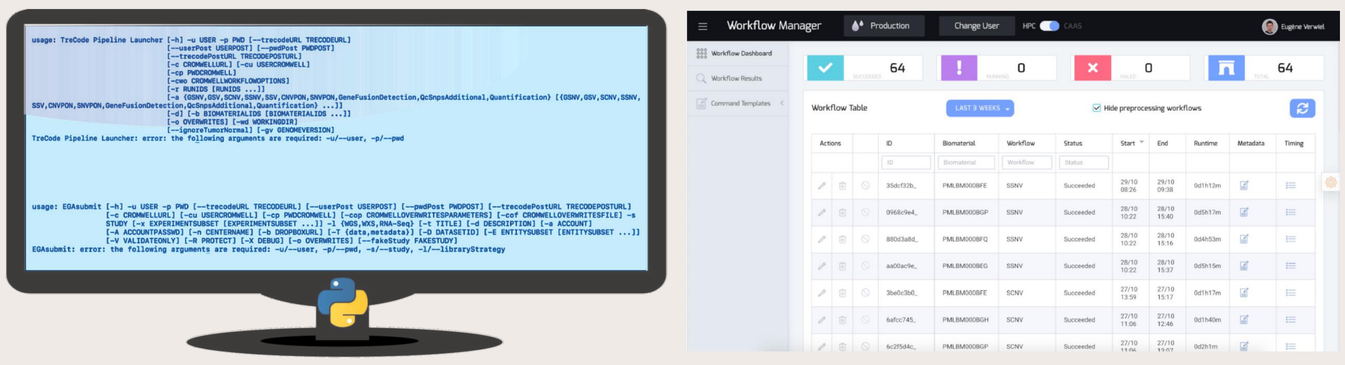
Method / Approach

1. A unifying data model has been developed from existing models in which various experimental setups and sequencing strategies can be described.
2. Model extensions have been added that describe the computational analysis as well as the sources consulted for this on the basis of dictionaries and ontologies, making results traceable, reproducible and FAIR.
3. The entire model became interoperable by loading it into Molgenis and to make the computational analysis code reusable on most compute backends, the workflow language WDL-Cromwell was implemented.
4. Client interfaces were built for automation.

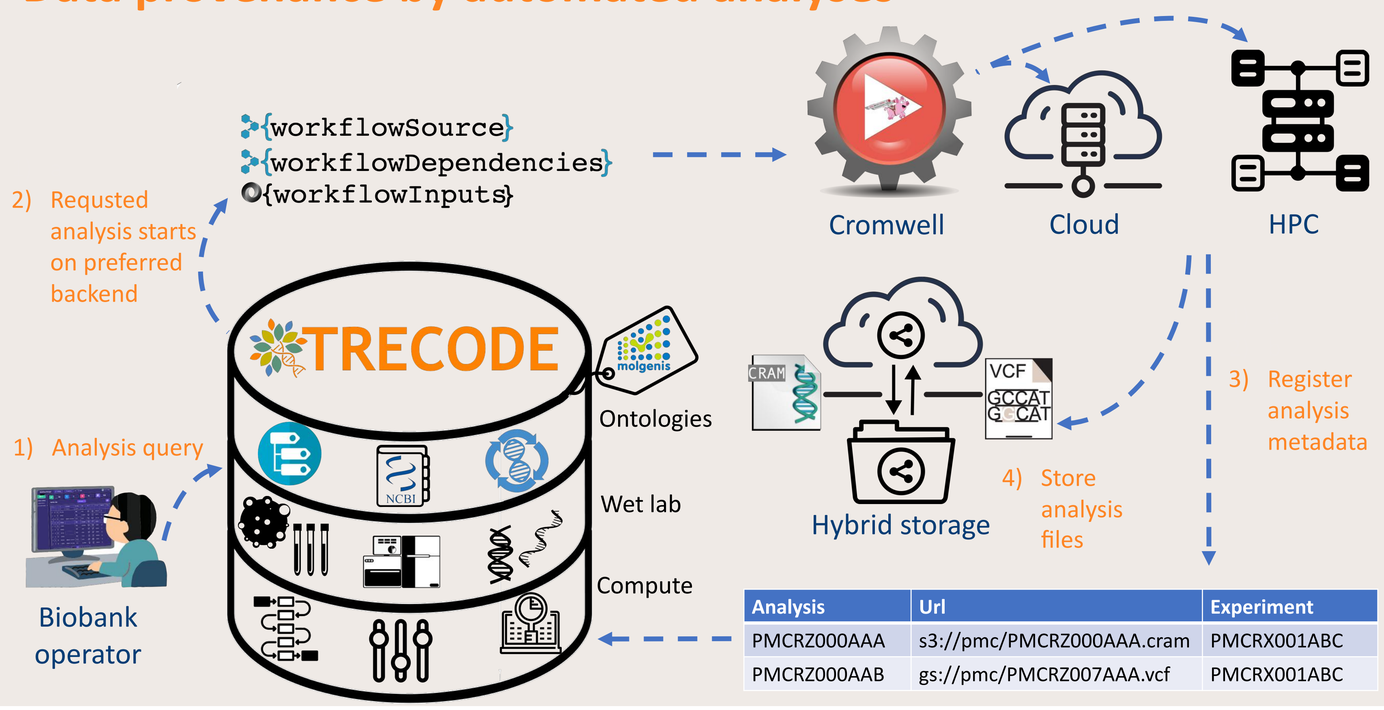
Features

1. Supports EGA submissions
2. Versatile command line client
3. Well arranged dashboard
4. Many re-usable components





Data provenance by automated analyses



Conclusion

- The Trecode platform demonstrates how FAIR data principles can be implemented in genomics with the principles of (meta)data harmonization, capture, integration and representation with which it facilitates large-scale data (re)use.
- Close integration of sample meta data with analysis (compute) data model supports automation of analysis and provides an audit trail of parameters used and analysis results.
- Strategies including the use of software containers, a workflow executor and an open data platform not only make the data but also the eco-system reusable.

References & acknowledgments

1. Kerstens, H. H. *et al.* Trecode: A FAIR Eco-System for the Analysis and Archiving of Omics Data in a Combined Diagnostic and Research Setting. *BioMedInformatics* **3**, 1–16 (2022).

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